THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — MATHEMATICS (Standard) · MID-TERM PRACTICE PAPER

Session 2026-27 · CBSE / Directorate of Education pattern · Sub code 041

Class: 10	Subject: Mathematics	Set 05	Time: 3 Hours	Max. Marks: 80

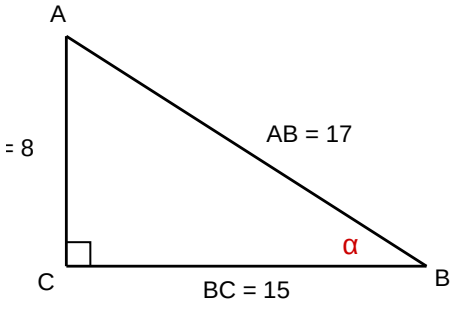
General Instructions:

1. This Question Paper has 5 Sections 'A', 'B', 'C', 'D' and 'E'.
2. Section A has 20 MCQs carrying 1 mark each (Q19–20 are Assertion–Reason questions).
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based questions (04 marks each) with sub-parts of the values of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice in 2 questions of 5 marks, 2 questions of 3 marks and 2 questions of 2 marks has been provided. An internal choice has been provided in the 2-mark questions of Section E. You have to attempt only one option out of the given options.
8. Draw neat figures wherever required. Take π = 22/7 wherever required, if not stated.
9. There is no provision of negative marking. Use of calculator is not permitted.
10. Please write down the serial number of the question before attempting it.

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — 20 questions · 20 marks

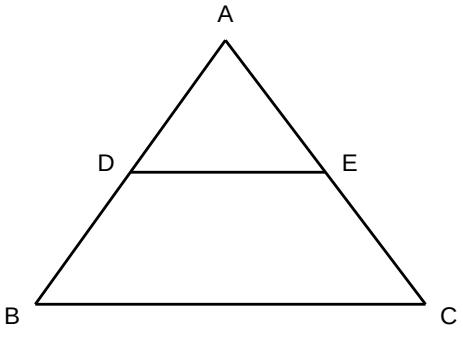
Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. From the given figure, the value of cos α is **1**



(a) $\frac{8}{17}$ (b) $\frac{8}{15}$ (c) $\frac{15}{17}$ (d) $\frac{15}{8}$

2. In the figure DE ∥ BC, D and E are the midpoints of AB and AC. If BC = 14 cm, then DE is **1**



(a) 14 cm (b) 7 cm (c) 28 cm (d) 3.5 cm

3. The HCF and LCM of 24, 36 and 60 are respectively **1**

(a) 6, 360 (b) 12, 720 (c) 24, 180 (d) 12, 360

4. The probability that a randomly chosen day of a non-leap year is in the month of April is **1**

(a) $\frac{30}{365}$ (b) $\frac{1}{12}$ (c) $\frac{31}{365}$ (d) $\frac{30}{366}$

5. Which of the following points CANNOT lie on the y-axis? **1**

(a) (0, −3) (b) (−3, 0) (c) (0, 0) (d) (0, 7)

6. The value of $(\sqrt{5} + \sqrt{2})^2 - 2\sqrt{10}$ is **1**

(a) $7 + 4\sqrt{10}$ (b) 3 (c) $5 + 2\sqrt{10}$ (d) 7

7. The number $5 \times 7 \times 11 + 7 \times 5$ is **1**

(a) prime (b) irrational

(c) composite (d) neither prime nor composite

8. The fares of a taxi are ₹ 30 for the first km and ₹ 12 for each additional km. The fare for 9 km is **1**

(a) ₹ 126 (b) ₹ 120 (c) ₹ 138 (d) ₹ 108

9. The pair $5x - 2y = 4$ and $10x - 4y = k$ is coincident for **1**

(a) k = 8 (b) k = 4 (c) k = 2 (d) k ≠ 8

10. The value of $(1 + \tan^2\theta)\cos^2\theta$ is **1**

(a) $\sin^2\theta$ (b) 1 (c) $\tan^2\theta$ (d) 0

11. The distance of the point (−6, 8) from the origin is **1**

(a) 14 (b) 2 (c) 10 (d) 6

12. A bag has 3 red, 5 white and 2 blue balls. One ball is drawn at random. The probability that it is not white is **1**

(a) $\frac{1}{5}$ (b) $\frac{3}{10}$ (c) $\frac{2}{5}$ (d) $\frac{1}{2}$

13. If $2x + y = 8$ and $x - y = 1$, then $x + y$ equals **1**

(a) 3 (b) 4 (c) 7 (d) 5

14. If the HCF of 45 and 75 is 15, then their LCM is **1**

(a) 225 (b) 150 (c) 450 (d) 1125

15. Two dice are thrown together. The probability that the sum is 7 is **1**

(a) $\frac{1}{12}$ (b) $\frac{5}{36}$ (c) $\frac{1}{6}$ (d) $\frac{7}{36}$

16. If $\tan\theta = \frac{5}{12}$, then $\sin\theta$ equals **1**

(a) $\frac{12}{13}$ (b) $\frac{5}{13}$ (c) $\frac{5}{12}$ (d) $\frac{13}{5}$

17. The equations $x + 2y = 4$ and $2x + 4y = 12$ have **1**

(a) one solution (b) two solutions

(c) infinitely many solutions (d) no solution

18. The common difference of the AP in which $a_{18} - a_{14} = 32$ is **1**

(a) 4 (b) 8 (c) −8 (d) −4

19. **Assertion (A):** Assertion (A): The mid-point of the segment joining (2, 5) and (6, 9) is (4, 7). **1**
Reason (R): Reason (R): The mid-point of (x_1, y_1) and (x_2, y_2) is $((x_1 + x_2)/2, (y_1 + y_2)/2)$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** Assertion (A): The probability of an impossible event is 0. **1**
Reason (R): Reason (R): The probability of a sure event is 1.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Find the HCF and LCM of 396 and 1080 by the prime factorisation method. **2**

OR

Three bells toll at intervals of 9, 12 and 15 minutes. If they toll together at 8 a.m., when will they next toll together?

22. A pole 6 m high casts a shadow 4 m long. At the same time a tower casts a shadow 50 m long. Find the height of the tower. **2**

23. In ΔABC, DE ∥ BC with AD = 4 cm, DB = 6 cm and AE = 6 cm. Find EC. **2**

OR

Two similar triangles have areas 49 cm² and 81 cm². If a side of the smaller is 14 cm, find the corresponding side of the larger.

24. A rope tied to the top of a vertical pole makes 30° with the level ground and is 24 m long. Find the height of the pole. **2**

25. Find the value of y if the distance between $(2, y)$ and $(10, -3)$ is 10 units. **2**

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that $5 - 3\sqrt{2}$ is irrational, given that $\sqrt{2}$ is irrational. **3**

27. Solve: $3x - 5y = -1$ and $x - y = -1$. **3**

28. Find the sum of all three-digit natural numbers which are multiples of 11. **3**

OR

The 9th term of an AP is 0. Prove that its 29th term is twice its 19th term.

29. Find the ratio in which the y-axis divides the line segment joining A(−4, −6) and B(10, 12). Also find the point of division. **3**

OR

The vertices of a triangle are (0, 0), (6, 0) and $(3, 3\sqrt{3})$. Show that it is equilateral.

30. If $\sin\theta = \frac{7}{25}$, find the value of $\sec\theta + \tan\theta$. **3**

31. A bag contains 4 red, 5 black and 6 green balls. A ball is drawn at random. Find the probability that it is (a) red (b) not green (c) red or black. **3**

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. **5**

33. A fraction becomes 9/11 if 2 is added to both numerator and denominator, and 5/6 if 3 is added to both. Find the fraction. **5**

OR

Draw the graphs of $2x + y = 6$ and $2x - y = 2$ and find the area of the triangle formed with the y-axis.

34. From the top of a 75 m high lighthouse the angles of depression of two ships on opposite sides are 30° and 45°. Find the distance between the ships. (Take $\sqrt{3} = 1.732$) **5**

35. Prove that $\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$. **5**

OR

If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, $0° < A + B \leq 90°$, $A > B$, find A and B.

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. A cricket academy holds a prize draw. 40 tickets numbered 1 to 40 are placed in a drum and one is drawn at random.

(i) Find the probability that the number is a multiple of 5. 1

(ii) Find the probability that the number is a perfect square. 1

(iii) Find the probability that the number is prime and odd. 2

OR

(iii) A ticket numbered 10 is drawn and kept aside. Find the probability that the next ticket drawn is even.

37. A farmer plots a field on a grid in which one unit is 10 m. The corners are A(1, 2), B(7, 2), C(7, 10) and D(1, 10).

(i) Find the length AB in metres. 1

(ii) Find the length of the diagonal AC in metres. 1

(iii) Find the coordinates of the point where the diagonals meet. 2

OR

(iii) A well is at the point P on AC with AP : PC = 3 : 2. Find P.

38. A cinema hall has 20 rows of seats. The first row has 15 seats and each next row has 2 more seats than the row before it.

(i) Write the first term and common difference of the AP. 1

(ii) How many seats are in the 10th row? 1

(iii) Find the total seats in the hall. 2

OR

(iii) Which row has 45 seats?